

VEDANG TRIVEDI

Software Engineer | Data Engineer | Distributed Systems, Python & ML Systems

✉ trivedivedang.02@gmail.com  [linkedin.com/in/vedang-trivedi-0389a91b9](https://www.linkedin.com/in/vedang-trivedi-0389a91b9)

 github.com/VEDANG2024  Ahmedabad, Gujarat, India

PROFILE SUMMARY

Backend and Software Engineer with expertise in Python development, distributed systems, PostgreSQL query optimization, and applied machine learning for systems efficiency. Designed, built, and shipped two enterprise applications at Kraft Heinz used daily by **50+ people** across **10 towers**, engineering automation estimated to save **1,600+ hours** a year. Also builds systems-level research: an entropy-aware LLM-routing model that cuts inference cost by **78%**, and a learned cardinality estimator for database query-plan optimization.

TECHNICAL SKILLS

Languages: Python, SQL, C++, C, JavaScript

Databases: PostgreSQL, Redis, MongoDB, MySQL

Systems & ML: Distributed Systems, Query Optimization, QLoRA/PEFT Fine-Tuning, XGBoost

App Dev & Automation: Git/GitHub, Power Apps, Power Automate, LangChain, Figma

Data & BI: Pandas/NumPy, Power BI, NLTK/spaCy

PROFESSIONAL EXPERIENCE

Kraft Heinz

KHMS/AIT Intern, GCC India

Feb 2026 – Present

Ahmedabad, Gujarat

- Engineered the AIT Power App end-to-end (5 screens, 20 SharePoint columns across 4 data types) after mapping and documenting the manual, email/Excel-based opportunity-tracking process, deploying a standardized 7-stage lifecycle model to **50+ users** across **10 towers**.
- Implemented 6+ conditional logic rules (field visibility, record locking, cascading filters, form validation) and 2 automated email-notification triggers, cutting an estimated **1,130+ hours** a year of manual status-update and reporting effort.
- Debugged and resolved **12 documented defects** through root-cause analysis across iterative build-test-fix cycles with real users, shipping the app entirely on existing Microsoft 365 licensing (Power Apps, SharePoint, Power Automate) with zero external budget.
- Migrated Health Check data onto a governed SharePoint backend (**5 lists**) and engineered the Health Check and OpEx Survey app suite – wizard-based flow, dependent dropdowns, submission validation – for **50+ users** across **11 towers**, cutting submission errors to near-zero and saving an estimated **267 hours** a year. Also implemented role-based access control across the platform, adding authentication and authorization rules with a limited-permission tier to enforce data governance without admin bottlenecks.

Karma Technolabs

Web Development Intern

Jan 2024 – Apr 2024

Ahmedabad, Gujarat

- Enhanced search, filtering, and data-upload backend functionality on a live CRM platform, contributing end-to-end across the full development, testing, and deployment lifecycle.

PROJECTS

UGAD-Lite: Uncertainty-Guided Adaptive Distillation | *ML Systems*



- Engineered an entropy-conformal router that dynamically directs queries between a large LLM (GPT-4o) and a fine-tuned Small Language Model (Phi-3-Mini via QLoRA), calibrated with conformal prediction for statistical safety guarantees.
- Cut LLM inference cost and token usage by 78% at a 94.8% task success rate on GSM8K and synthetic arithmetic benchmarks, with a <5% false-negative rate.

CardEst | *SQL, Query Optimization*



- Built an XGBoost-based cardinality estimator trained on the TPC-H dataset to improve database query-optimizer execution-plan accuracy in PostgreSQL and relational systems.
- Cut query latency and resource usage in 70% of evaluated cases by improving Pearson correlation and reducing median estimation error.

Raft Census (DBFS) | *Distributed Systems, Raft Consensus*



- Built a language-agnostic, decentralized file-storage system using Raft cluster consensus (Redis + MongoDB-backed), with work-stealing task distribution to support concurrent multi-client create/read/write/delete operations with efficient search, caching, and asynchronous task execution.

UML2PlantUML | *LLMs, Machine Learning*



- Built a small-scale, local reproduction of the approach from Sadeghi et al. (MODELS 2026, arXiv:2607.28987) for handwritten UML-to-PlantUML generation, preprocessing a dataset of activity and sequence diagram images for model training.
- Fine-tuned and benchmarked 7 vision-language models (ViT-BART, BLIP, Flan-T5, ViT-GPT2, Pix2Struct, CLIP, MiniCPM) to generate PlantUML code from class diagram images, and deployed the trained model on Kaggle, where it has drawn numerous downloads.

Distributed Database System | *2PC, Query Optimization*



- Engineered a distributed database with a coordinator-worker topology, implementing horizontal/vertical data fragmentation and a global data dictionary for location-transparent query routing.
- Implemented Two-Phase Commit (2PC) for multi-node transaction atomicity and semi-join query optimization, cutting cross-node bandwidth by 60–70% and query latency by 3–4x versus centralized execution.

EDUCATION

DA-IICT (Dhirubhai Ambani Institute of Information and Communication Technology)

M.Tech, ICT (Software Systems) | CPI: 7.47/10

Gujarat Technological University

B.E., Computer Engineering | CGPA: 7.77/10

2024 – 2026

Gandhinagar, Gujarat

2020 – 2024

Ahmedabad, Gujarat

POSITIONS OF RESPONSIBILITY

Teaching Assistant – Software Engineering & Object-Oriented Programming

DA-IICT, Gandhinagar

Aug 2024 – Present

Gandhinagar, Gujarat

- Conducted labs and doubt-clearing sessions for 100+ students, designed technical documentation, assignments, and exams, and mentored students in Software Engineering and Object-Oriented Programming.

ACHIEVEMENTS

- GATE CS 2024 qualifier.